

Elias-Fano coding

Advanced Data Structures – Summer 2026 – Lecture 2

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Previous slides: Bit vectors, Rank & Select

- Bit vector: storing bits
- Rank: counting 1 bits
- Select: finding 1 bits

Goal: Storing lists of integers

How do we efficiently store a list of integers

$$0 \leq x_0 \leq x_1 \leq \cdots \leq x_{n-1} < U.$$

Theorem 1 Lower bound.

Using *stars and bars*, there are $\binom{U+n-1}{n}$ of choosing n elements (with duplicates) from $\{0, 1, \dots, U-1\}$.

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For $n = o(\sqrt{U})$ and $n \rightarrow \infty$,

$$\frac{1}{n} \log_2 \binom{U+n-1}{n} \rightarrow \log_2 \left(e \cdot \frac{U}{n} \right).$$

Storing many small integers ($n \gg U$)

Question 1.

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Storing dense lists ($n \approx U$)

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- $U + n$ bits

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- Lower bound for $n = u$:

$$\bullet U + n \text{ bits} \quad \log_2 \binom{U + n - 1}{n} = \log_2 \binom{2n - 1}{n} = \log_2 (\Theta(2^{2n} / \sqrt{n})) \approx 2n$$

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Consecutive elements share the $\approx \log_2 n$ high bits!

Elias-Fano coding (Elias 1974; Fano 1971; Vigna 2013)

Theorem 2 Elias-Fano (EF) coding.

A list of n integers $0 \leq x_0 \leq x_1 \leq \dots \leq x_{n-1} < U$ can be stored in

$$n(\log_2(U/n) + 2)$$

bits while supporting $O(1)$ access time.

This is only $2 - \log_2(e) = 0.5573$ bits/key away from the lower bound when $n = o(\sqrt{U})$!

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- We have budget to store the $\ell = \lfloor \log(U/n) \rfloor$ bits explicitly.
- But how do we encode all $\log U - \ell \approx \log n$ high bits using ≈ 2 bits/key?
- Using the $n \approx U$ case!

Elias-Fano coding

Definition 1.

Split each integer into a $\ell = \lfloor \log(U/n) \rfloor$ -bit *low* part and $(\log U - \ell)$ -bit *high* part.

- Store all low bits explicitly in $n\ell$ bits.

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- Write how often each high part occurs in unary.
 - $\lceil U/2^\ell \rceil$ high parts with

$$n \leq \lceil U/2^\ell \rceil < 2n.$$

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- In total $n + \lceil U/2^\ell \rceil \leq 3n$ bits.

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- Store all low bits explicitly in $n\ell$ bits.
- Thus, space usage below $n\ell + 3n = n(\lfloor \log(U/n) \rfloor + 3)$
- More refined analysis shows that actually:
 $n\lfloor \log(U/n) \rfloor + n + \lceil U/2^\ell \rceil \leq n(\log(U/n) + 2).$
- In total $n + \lceil U/2^\ell \rceil \leq 3n$ bits.

Interpretations of the high-bit encoding

- Each input number corresponds to a 1.
- Each "bucket" for the high parts corresponds to a 0.
- 0 separates the unary count in number of 1 for each high part.
- The number of 0 directly before a 1 is the jump the high part makes from the previous value.
- The total number of 0s before a 1 is the value of the high part.

Querying Elias-Fano

Definition 2.

To get x_i :

- Read the i 'th group of ℓ low bits.

Querying Elias-Fano

Definition 2.

To get x_i :

- Get the high part of the i 'th value:
 - Count zeros before the i 'th 1 (counting from 0):
$$\text{high}(i) = \text{select}_1(i) - i$$
- Read the i 'th group of ℓ low bits.

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- For *strictly increasing* integers: replace x_i by $x_i - i$.
- To store a non-sorted list with small sum: store prefix sums.

Further reading / sources

- Giulio Pibiri's [notes on Elias-Fano](#)
- Modern paper reviving/applying it: (Vigna 2013)
- Original papers: (Elias 1974; Fano 1971)
- Dynamic version: (Pibiri and Venturini 2017)
- `sux::EliasFano` [implementation](#)

Bibliography

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Possible exam questions

- What is the core idea of Elias-Fano coding?
- What is the stars & bars technique? Relate it to a binomial coefficient.
- How many "high" bits are there? Why does this make sense?
- What is the space of the Elias-Fano encoding? Is it far from the lower bound?
- What are some interpretations of the bits encoding the high parts?
- How does one recover x_i , i.e., query the Elias-Fano coded list?

Blackboard: Bit vector & Rank

Elia-Fano
Lower bound.

Stars & Bars $n=8$ $u=4$

$x_1 x_2 x_3 x_4$ | $x_5 x_6$ | $x_7 x_8$
 B_0 B_1 B_2 B_3

$0, 0, 0, 0, 2, 2, 3, 3$

$x_1 x_2 x_3 x_4$ | $x_5 x_6$ | $x_7 x_8$ = 102
 $x_1 x_2$ | x_3 | $x_4 x_5$ | $x_6 x_7$ = 101

$x_{i,t} \rightarrow x_{i,t+1}$

0 1 2 3 6 7 9 10
 1 1 1 0 0 1 1 0 1 1 0

$\binom{n+u-1}{n}$
 $\binom{2n}{n}$

$\approx N(\frac{n}{2}, \sigma)$
 $\sigma^2 = \frac{n}{4}$
 $\sigma = \frac{\sqrt{n}}{2}$
 $O(\sqrt{n})$

$\sum_{i=0}^n \binom{2n}{i} = 2^{2n}$

Blackboard: Select & Darray

x_0

	H_i	L_i	H_i
1	0 000001		0
5	0 000101		1
30	0 011110		2
32	0 1000000		2
33	0 1000001		2
66	1 000010		4
67	1 000011		4
68	1 000100		4
96	1 1000000		6
100	1 1000100		6

$+2$

1 1 0 1 0 1 1 0 0 1 1 1 0 0 1 1 0

0 1 2 3 4 5 6

$$select_1(5) = 9 - 5 = 4$$

How many 0 before select(j)?

$$rank_0(select_1(j)) =$$

$$select_1(j) - j$$

$$0 \leq i \leq 2^{\lceil \lg n \rceil} = U \approx n$$

Stars & Bars Plain array

SUX